RiskDrive: Multimodal-Based Risk Map Prediction for Autonomous Driving

Runhao Li
1006746309, rainli@mail.utoronto.ca
https://github.com/rainli10/riskmap

1 Project introduction

Autonomous driving scenes are visually complex and contain many objects, but not all regions are equally important for safe navigation. For example, a nearby pedestrian or vehicle is usually much more critical than a building, sidewalk, or distant background region. Our project is motivated by the idea that an autonomous system should not only understand what is in the scene, but also estimate how risky each region is to the ego vehicle.

The goal of this project is to predict a dense per-pixel risk map from multimodal visual input. The model takes an RGB image together with a depth map as input, and outputs a 5-class dense risk prediction over the full image. Higher predicted risk values correspond to scene regions that are more dangerous for driving. To build supervision for this task, we construct target risk maps by combining two factors: a fixed semantic risk weight for each object category, and a depth-based weight that increases as objects become closer to the ego vehicle. For each segmented object region, we use its minimum depth so that the closest visible part of the object determines its danger level.

This task goes beyond standard semantic segmentation by requiring the model to jointly reason about semantic meaning and geometric proximity for risk prediction. The model must learn not only which objects are more dangerous in a road scene but also how their distance from the ego vehicle changes the final risk level. This making this a more challenging task than pure segmentation alone. Such a dense risk map could be useful for downstream planning, decision making, or safety monitoring in autonomous driving. To study this problem, I compare two deep learning approaches: a simple CNN trained from scratch as a baseline, and a stronger SegFormer-based model adapted for RGB-D input. This comparison helps evaluate whether lightweight convolutional features are sufficient for dense risk prediction, or whether a pretrained transformer-based architecture provides a clear advantage.

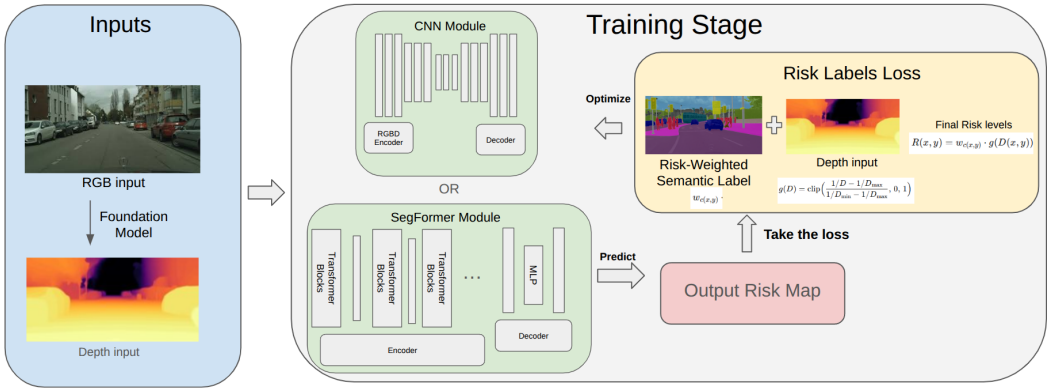

Figure 1: The proposed risk map prediction framework

2 Notable Contribution

2.1 Data Processing

I built the risk-map dataset from the Cityscapes [1] driving dataset by using RGB image together with its semantic segmentation map and a generated depth map from DepthAnything Model [3]. I constructed a new supervision target by combining semantic importance and geometric proximity. Specifically, each semantic class is assigned a fixed risk weight w_c based on its expected danger level in road scenes, for example, pedestrians get higher weight than cars than static buildings. For each segmented object region we use its minimum depth to represent how close that object is to the ego vehicle. We then convert depth into a normalized proximity weight using

$$g(D) = \text{clip}\left(\frac{1/D - 1/D_{\max}}{1/D_{\max} - 1/D_{\min}},\ 0,\ 1\right)$$

so that closer objects receive higher values. The final dense risk map used as supervision is computed per pixel by

$$R(x, y) = w_{c(x,y)} \cdot g(D(x, y))$$

The example of this risk map is showing in Figure 2. This dataset design was one of the main challenges of the project, since risk map prediction is not a standard benchmark task, so we had to define the labeling logic ourselves based on everyday driving intuition, including the semantic weights, inverse-depth weighting, and overall risk construction process.

(a) RGB (b) Semantics (c) Risk Map Discrete (d) Risk Map Continuous

Figure 2: Example training sample used in our dataset construction.

2.2 Baseline Model

As a baseline, we use a simple CNN trained from scratch for dense per-pixel risk classification. The model takes a 4-channel input consisting of RGB-D. It outputs a 5-class (from low risk to high risk) prediction map with the same spatial resolution as the input. The baseline architecture is a fully convolutional CNN that takes a 4-channel RGB-D input and passes it through six Conv-BatchNorm-ReLU blocks with channel sizes $32 \to 64 \to 128 \to 256 \to 64 \to 32$, followed by a final 1×1 convolution layer to produce the dense risk prediction map.

The baseline is trained using class-weighted cross-entropy loss, where the class weights help reduce the imbalance between common low-risk pixels and much rarer high-risk pixels. This is important because most pixels in driving scenes belong to background or low-risk regions, while truly dangerous areas occupy a much smaller part of the image. Quantitatively, the baseline is evaluated using validation loss, pixel accuracy and MAE (Mean Absolute Error) in risk space. Our best baseline result so far achieved a validation loss of **0.0933**, pixel accuracy of **0.809**, MAE of **0.064**.

Qualitatively, as shown in Figure 3, the baseline model is able to identify major high-risk objects such as cars and pedestrians, which suggests that it has learned the general structure of the task. However, the predicted object regions are often blurry and less precise around boundaries, especially for smaller or more distant objects. The output also appears to rely heavily on image structure such as edges and local high-frequency details, sometimes treating visually textured regions as a whole object without clearly distinguishing finer risk differences within the scene. A likely reason is that the simple CNN mainly learns local convolutional features and has weaker global scene understanding compared with stronger architectures. Still, for a lightweight model trained from scratch, the result is decent and serves as a reasonable baseline for comparison with our primary model.

(a) RGB/Overlay/Predicted (b) GT Risk Map

Figure 3: Example of prediction result and comparison with GT.

Table 1: Comparison between the baseline CNN and our SegFormer-based model on the validation set.

Model	Validation Loss ↓	Pixel Accuracy ↑	MAE in Risk Space ↓
Baseline CNN	0.0933	0.809	0.0640
Primary Model	0.0792	0.891	0.0453

2.3 Primary Model

Our primary model is a SegFormer-based [2] transformer network for 5-class per-pixel risk classification from RGB-D input. Since pretrained SegFormer expects 3-channel input, we first use a learned 1×1 convolution to map the 4-channel RGB-D input to 3 channels. The adapted input is then processed by a pretrained SegFormer backbone, where a hierarchical transformer encoder extracts multi-scale features through overlapping patch embeddings and transformer blocks, and a lightweight MLP decoder fuses and upsamples these features to produce the final dense risk map. We chose this architecture because risk prediction requires both semantic understanding and geometric reasoning. Compared with the baseline CNN, transformer provides stronger feature extraction and better global scene understanding, which helps identify important objects such as cars and pedestrians while also capturing their surrounding context. In addition, pretrained Cityscapes features make it a strong match for urban driving scenes and a better starting point than training a simple CNN from scratch.

I design the training strategies to train the SegFormer-based model in two stages. In **Stage 1** (epochs 1–8), the pretrained backbone is frozen and only the input adapter and decode head are trained using AdamW with cosine annealing. In **Stage 2** (epochs 9–30), the backbone is unfrozen and the full model is fine-tuned with a smaller learning rate for the pretrained backbone and a larger learning rate for the newly added layers, again using AdamW with cosine annealing. In both stages, gradient clipping and automatic mixed precision on CUDA are used for stable and efficient training. Quantitatively, I evaluate the model using the same metric showing in Table 1. The best primary-model result so far achieved a validation loss of **0.0792**, pixel accuracy of **0.891**, and MAE of **0.0453**. Qualitatively, the model produces cleaner and more coherent risk regions than the baseline with better object separation and sharper boundaries. It also handles human regions much better, as shown in Figure 4. However, the predicted boundaries are still not fully precise, and some object edges remain blurry.

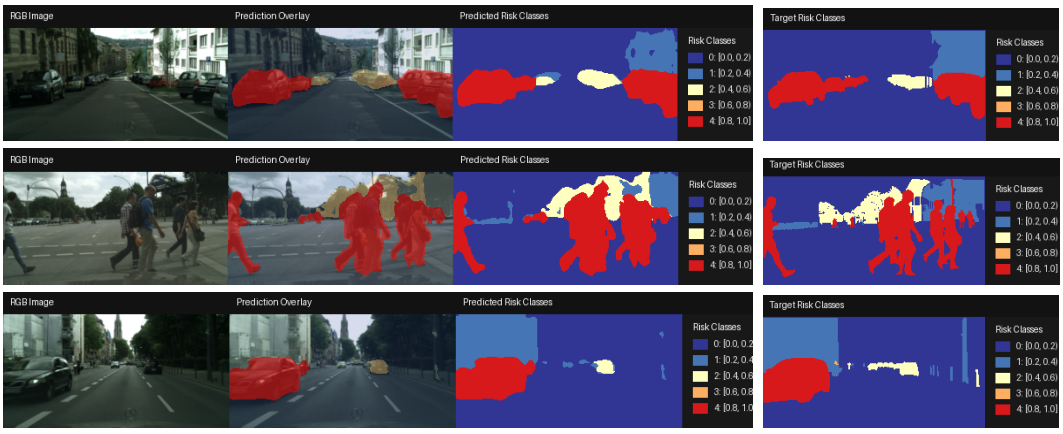

Figure 4: Qualitative comparison of predicted risk maps and ground-truth risk maps.

References

[1] Marius Cordts et al. *The Cityscapes Dataset for Semantic Urban Scene Understanding*. 2016. arXiv: 1604.01685 [cs.CV]. URL: https://arxiv.org/abs/1604.01685.

[2] Enze Xie et al. *SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers*. 2021. arXiv: 2105.15203 [cs.CV]. URL: https://arxiv.org/abs/2105.15203.

[3] Lihe Yang et al. *Depth Anything: Unleashing the Power of Large-Scale Unlabeled Data*. 2024. arXiv: 2401.10891 [cs.CV]. URL: https://arxiv.org/abs/2401.10891.